Properties of the Exponential Function and Their Role in Machine Learning

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Abstract—*The exponential function exp(x) is the single most frequently occurring function in machine learning theory. This paper gives a rigorous account of its series definition, its convergence properties, and the algebraic identities — homomorphism, strict positivity, self-derivative, monotonicity, and convexity — that follow from that definition. It then traces how the fact that the exponential series converges absolutely for every real (and complex) argument is what makes moment generating functions, the Chernoff method, and concentration inequalities universally applicable in learning theory, and how the same properties make the softmax function — together with the log-sum-exp trick — both mathematically well-behaved and numerically stable in neural networks. The exponential family of distributions and its connection to the Boltzmann distribution are discussed as a further consequence of the same algebraic structure.*

Index Terms—*exponential function, convergence, moment generating functions, Chernoff bound, softmax, exponential families*

I. Introduction

The exponential function is the backbone of a disproportionate share of machine learning theory. It appears in the softmax layer that turns network outputs into probabilities, in the moment generating functions (MGFs) that drive concentration inequalities, in the Boltzmann distribution that underlies energy-based models, and in the exponential family of distributions that includes the Gaussian, Bernoulli, and Poisson distributions as special cases. All of these constructions rely on the same small set of algebraic properties of exp, which in turn all follow from a single analytic fact: the defining power series of exp converges absolutely for every real number.

II. The Exponential Series and Its Convergence

The exponential function is defined by the power series

$$e^x \;=\; \sum_{n=0}^{\infty} \frac{x^n}{n!} \;=\; 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

This series converges absolutely for every $x \in \mathbb{R}$, and in fact for every $z \in \mathbb{C}$. The reason is that the factorial n! in the denominator grows faster than any fixed power x^n in the numerator, so for large n the ratio of consecutive terms $|x|/(n+1)$ tends to zero regardless of the size of x; the ratio test then guarantees absolute convergence. Differentiating the series term by term and multiplying two such series using the binomial identity yields the two defining properties

$$\frac{d}{dx} e^x = e^x, \qquad e^0 = 1, \qquad e^{a+b} = e^a e^b$$

Absolute convergence for every x means that exp is defined and well-behaved everywhere, with no domain restriction of any kind. This single fact is what allows an expression such as e^{tX} to be written for any random variable X and any real t without first checking that the expression is well-defined — a convenience that later sections show is essential to the Chernoff method.

III. Algebraic and Analytic Properties

Four properties of exp, all consequences of the series definition, account for essentially every appearance of the exponential in machine learning theory.

A. Homomorphism

The identity $e^{a+b} = e^a \cdot e^b$ turns addition into multiplication. This is why the moment generating function of a sum of independent random variables factors as a product of individual MGFs, and why shifting every logit in a softmax by the same constant leaves the output distribution unchanged.

B. Strict Positivity

$e^x > 0$ for every $x \in \mathbb{R}$; the exponential never touches zero. This guarantees that e^{tX} is always a valid non-negative random variable, which is exactly the condition Markov's inequality requires, and it guarantees that softmax outputs are always strictly positive probabilities.

C. Self-Derivative and Uniqueness

The property $d/dx\, e^x = e^x$ characterizes exp uniquely up to a scalar multiple: if f is differentiable, $f'(x) = f(x)$ for all x, and $f(0) = 1$, then $f(x) = e^x$. The proof considers $g(x) = f(x)e^{-x}$; the quotient rule gives $g'(x) = 0$, so g is constant, and $g(0) = 1$ forces $f(x) = e^x$ for all x. Consequently, any continuous function satisfying $f(a + b) = f(a)f(b)$ with $f(0) = 1$ must equal e^{cx} for some constant c — the exponential is not a convenient choice for building such functions, it is the only one.

D. Monotonicity and Convexity

exp is strictly increasing, and it is convex: $e^{\lambda a + (1-\lambda)b} \le \lambda e^a + (1-\lambda)e^b$ for $\lambda \in [0,1]$. Convexity, via Jensen's inequality, gives $e^{E[X]} \le E[e^X]$ for any random variable X, a bound used repeatedly in the derivation of moment generating function inequalities.

IV. Taylor Remainder and Approximation Bounds

Truncating the exponential series after n terms gives the Taylor polynomial $T_n(x) = \Sigma(k=0 \text{ to } n)\ x^k/k!$. For $x \geq 0$ the Lagrange remainder gives

$$0 \;\leq\; e^x - T_n(x) \;\leq\; \frac{x^{n+1}}{(n+1)!}\, e^x$$

and for $x \leq 0$ the series alternates in sign, giving $|e^x - T_n(x)| \leq |x|^{n+1}/(n+1)!$. Two low-order instances of this bound are used throughout probability and learning theory: the first-order bound $e^x \geq 1 + x$ holds for every $x \in \mathbb{R}$ with equality only at $x = 0$, and the second-order bound $e^x \leq 1 + x + x^2/2$ holds for $x \leq 0$. The latter is the key algebraic step in the proof of Hoeffding's lemma.

V. Convergence and Its Effect on Machine Learning: MGFs and the Chernoff Method

The moment generating function

$$M_X(t) \;=\; \mathbb{E}\big[e^{tX}\big]$$

exploits every property of exp discussed above simultaneously: positivity keeps $M_X(t) > 0$ so that Markov's inequality applies; the homomorphism property gives $M_{X+Y}(t) = M_X(t)M_Y(t)$ for independent X and Y, turning sums of random variables into products of MGFs; the self-derivative property encodes moments as derivatives, $d^n/dt^n\ M_X(t)|_{t=0} = E[X^n]$; and convexity gives the Jensen bound $e^{E[tX]} \leq M_X(t)$.

The Chernoff method applies Markov's inequality to e^{tX}:

$$P(X \geq a) \;=\; P\big(e^{tX} \geq e^{ta}\big) \;\leq\; \frac{\mathbb{E}[e^{tX}]}{e^{ta}} \;=\; M_X(t)\, e^{-ta}$$

Optimizing over $t > 0$ gives the tightest such bound. Because the exponential series converges absolutely for every real argument, this construction never runs into a domain issue — e^{tX} is always well-defined, and the resulting bound is either informative or, in the worst case, trivially $+\infty$ for heavy-tailed distributions whose MGF diverges. It is convergence, not boundedness, that the method requires. For a bounded, zero-mean random variable X with $|X| \leq c$, applying the second-order bound $e^x \leq 1 + x + x^2/2$ inside the expectation yields the sub-Gaussian estimate $E[e^{tX}] \leq e^{t^2c^2/2}$, the key lemma behind Hoeffding's inequality — a bound used throughout statistical learning theory to control the deviation of empirical averages from their expectation.

VI. The Exponential in Softmax and Numerical Stability

The softmax function converts a vector of raw scores into a probability distribution:

$$p_i \;=\; \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Three properties of exp make this construction work. Strict positivity guarantees every p_i is a valid, strictly positive probability. The homomorphism property means that adding the same constant c to every logit leaves the output unchanged, since $e^{z\_i+c}/\Sigma_j\ e^{z\_j+c} = e^{z\_i}/\Sigma_j\ e^{z\_j}$; this shift-invariance is exactly what justifies the log-sum-exp trick. Monotonicity ensures that larger logits map to larger probabilities, preserving the ranking of the inputs.

VII. Log-Sum-Exp and Exponential Families

The log-sum-exp function

$$\mathrm{LSE}(z) \;=\; \log\Big(\sum_i e^{z_i}\Big)$$

is convex, and its gradient is exactly the softmax function, $\partial\mathrm{LSE}(z)/\partial z_i = e^{z\_i}/\Sigma_j\ e^{z\_j}$. It is also a smooth approximation to the maximum, $\max_i z_i \leq \mathrm{LSE}(z) \leq \max_i z_i + \log n$, tight when a single component dominates.

A worked numerical example illustrates why this matters in practice. For logits z = [1000, 1001, 999], computing e^{1000} directly overflows floating-point arithmetic. Subtracting c = max(z) = 1001 before exponentiating gives $e^{-1} \approx 0.368$, $e^0 = 1$, and $e^{-2} \approx 0.135$, which sum to 1.503 and yield the softmax output [0.245, 0.665, 0.090]. Every intermediate value now lies between 0 and 1, and the shift-invariance property guarantees this is exactly the result that direct computation would have produced had it not overflowed. Every practical implementation of softmax relies on this trick, and it exists only because of the homomorphism property of exp.

The same algebraic structure underlies the exponential family of probability distributions, whose densities take the form

$$p(x \mid \theta) \;=\; h(x)\, \exp\big(\theta^\top T(x) - A(\theta)\big)$$

The Gaussian, Bernoulli, Poisson, and most other standard distributions are exponential families, and the Boltzmann distribution $p(x) \propto e\hat{}(-E(x)/T)$ that underlies energy-based models is a direct instance of the same form, connecting statistical learning to statistical mechanics.

VIII. Two Common Pitfalls

Two subtleties are worth flagging explicitly. First, e^x is convex while log x is concave; these are dual facts, giving the two directions of Jensen's inequality, $e^{E[X]} \leq E[e^X]$ and $E[\log X] \leq \log E[X]$, which are used respectively in MGF bounds and in proving that Kullback–Leibler divergence is non-negative.

Second, the homomorphism property is a genuinely scalar phenomenon. For square matrices A and B, the matrix exponential converges for every A, and the product identity holds only when A and B commute:

$$e^A = \sum_n \frac{A^n}{n!}, \qquad e^{A+B} = e^A e^B \;\; (\text{iff } AB = BA)$$

In general the correct identity is given by the Baker–Campbell–Hausdorff formula

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\cdots}$$

where [A,B] = AB − BA. This distinction matters whenever exponential-family or Lie-group structure is used in higher-dimensional model classes, where the convenient scalar identities of Section III no longer hold without qualification.

IX. Conclusion

Every major appearance of the exponential function in machine learning theory — softmax, moment generating functions, the Chernoff method, concentration inequalities, and exponential family models — traces back to the same short list of algebraic properties: homomorphism, strict positivity, self-derivative, monotonicity, and convexity. All of these in turn rest on a single analytic fact, that the defining series of exp converges absolutely for every real argument. It is this convergence property, more than any single identity, that makes the exponential function usable without qualification anywhere in learning theory: the resulting bounds and constructions may occasionally be loose or uninformative, but they are never undefined.

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